

DNA Methylation Age Prediction with Multi-Field Tokenization

1 Introduction

DNA methylation patterns change systematically with age, making them powerful biomarkers for biological age prediction. However, methylation data poses challenges: high dimensionality (thousands of CpG sites), complex inter-site correlations, and continuous values in $[0,1]$. Traditional approaches use linear models (e.g., ElasticNet), but recent work shows that Transformer-based models can capture non-linear dependencies between CpG sites.

We base our work on MethylGPT [1], a foundation model that treats CpG sites as tokens and applies self-attention to learn methylation representations. We propose adapting the BMFM-RNA architecture [2], originally designed for single-cell RNA-seq, to methylation data. Our hypothesis is that BMFM-RNA’s multi-field tokenization—which separately encodes site identities and continuous values—may better preserve fine-grained methylation information compared to MethylGPT’s approach.

- **Article paper: MethylGPT (bioRxiv 2024)**
- **BiomedSciAI ACL Project (GitHub)**

2 Data Description

Table 1: Dataset summary and preprocessing.

Item	Description
Pretraining sources	EWAS Data Hub; ClockBase
Collected profiles	226,555
After QC + deduplication	154,063
Platforms	Illumina 27K / 450K / EPIC
CpG harmonization	49,156 sites (≥ 5 EWAS traits and/or $\geq 95\%$ presence)
Value range	β -values in $[0,1]$
Missingness handling	Retained for masking-based objectives

3 Baseline Architecture

Pretraining data. Large-scale methylation profiles (154,063 after QC) are used to learn general representations before downstream fine-tuning.

This model fits our data because methylation profiles contain structured CpG dependencies that benefit from self-attention, while pretraining on large-scale methylation data learns generalizable token representations.

Table 2: Architecture comparison (pretraining vs fine-tuning).

Component	Pretraining	Fine-tuning
Base model	TransformerModel (scGPT)	Same TransformerModel (frozen/low LR)
Encoder	CpG embeddings	Same
Value encoder	ContinuousValueEncoder	Same
Transformer	6 layers, 4 heads, $d = 64$	Same (loaded from pretrained)
Task head	ExprDecoder (MLM) + MVCDecoder	ResNet1D age head
Output	Methylation reconstruction	Single age prediction
Dropout	0.1	0.0
Mask ratio	30%	0%

The pretrained Transformer transfers to the 8K-CpG setting, and the ResNet1D head maps learned embeddings to age.

Fine-tuning data. We fine-tune on 11,453 labeled samples, each represented by 8,000 CpGs per sample ($M=8000$), with ages spanning 0–100 years and multi-tissue composition (blood $\sim 47.2\%$, brain $\sim 34.5\%$, others $\sim 18.3\%$).

Fine-tuning task. Predict chronological age from the Transformer embeddings using a ResNet1D regression head (Conv1D 3x3, 6x ResBlock1D, AdaptiveAvgPool1d, Dropout, Linear, Sigmoid) with MSE loss. The output is a normalized age in $[0, 1]$. The baseline results are reported in Section 6.

Architecture flow: Pretraining applies MLM-style masked value prediction using the ExprDecoder; fine-tuning replaces the decoder with a ResNet1D age head that operates on Transformer outputs (excluding [CLS]) to produce a single normalized age prediction.

4 Loss Function

Pretraining uses masked MSE to reconstruct methylation values (MLM) with a 30% mask ratio, along with a profile-level reconstruction objective. Fine-tuning uses MSE on normalized chronological age with no masking.

5 Evaluation Plan

We use a fixed 48/12/40 train/val/test split. Model selection uses validation MAE. The test set is used once per run for final evaluation. Generalization is assessed by repeating training with multiple random seeds (40–44). Metrics reported for age prediction include MAE (interpretable in years), MedAE (robust to outliers), RMSE, R^2 (proportion of variance explained), Pearson r , and Spearman r . These are standard metrics for age regression that allow comparison with prior work [3, 4].

6 Baseline Results

Test metrics are computed from the best checkpoint (selected by validation MAE). Validation MAE is averaged across all 100 training epochs. For visual tracking and seed-level curves, see:

- **MethylGPT W&B Results**

Table 3: MethylGPT baseline results (48/12/40 split, 5 seeds).

Metric	Mean $\pm$ Std
Validation MAE (years)	5.12 ± 0.89
Test MAE (years)	4.95
Test MedAE (years)	3.06 ± 0.20
Test R^2	0.911
Test Spearman	0.958

7 Proposed Changes: BMFM-RNA Architecture

We adapt the BMFM-RNA SCBert architecture for methylation-based age prediction. The original BMFM-RNA is a BERT-based foundation model for single-cell RNA-seq (~ 110 M parameters). We scale it down to match our 8K-CpG input.

- **Reference: BMFM-RNA (IBM Research Project)**
- **Code: GitHub Repository**

7.1 Why BMFM-RNA for Methylation?

Multi-field tokenization. Unlike MethylGPT which combines site and value information implicitly, BMFM-RNA uses explicit dual-field representation:

$$\mathbf{h}_i = \mathbf{e}_{\text{cpg}}(s_i) + \mathbf{e}_{\beta}(\beta_i) \quad (1)$$

where $\mathbf{e}_{\text{cpg}}$ captures site-specific information (genomic context) and $\mathbf{e}_{\beta}$ encodes the methylation level via an MLP. This separation allows the model to learn which sites are informative independently of their current value.

Continuous value encoding. The β -value encoder projects continuous values into embedding space without discretization, preserving fine-grained information.

Larger capacity. BMFM-RNA uses 512-dimensional embeddings and 8 attention heads (vs. MethylGPT’s 64-dim, 4 heads), potentially capturing more complex CpG interactions.

7.2 Tokenizer and Multi-Field Representation

We adapt BMFM-RNA’s `MultiFieldTokenizer` for methylation. The vocabulary contains 5 special tokens ([UNK], [SEP], [PAD], [CLS], [MASK]) plus 8,000 CpG site tokens (total: 8,005).

Two-field input. Each sample is represented as two parallel sequences:

- **cpg_sites:** [[CLS], cg00000029, cg00000108, ..., [PAD], [PAD]]
- **beta_values:** [0, 0.85, 0.12, ..., 0, 0]

The CpG ID embedding captures site identity; the β -value encoder captures methylation state.

7.3 Architecture Comparison

Table 4: Architecture comparison: original BMFM-RNA, our adapted version, and MethylGPT.

Parameter	BMFM-RNA (original)	BMFM-RNA (ours)	MethylGPT (scGPT)
Layers	12	6	6
Attention heads	12	8	4
Hidden size d	768	512	64
FFN size	3072	2048	256
Head dimension	64	64	16
Parameters (encoder)	$\sim 110\text{M}$	$\sim 23\text{M}$	$\sim 2\text{M}$
CpG subset	–	8,000	8,000

7.4 Pretraining Results

Before fine-tuning, we pretrain the BMFM-RNA encoder using masked language modeling (MLM) on methylation β -values. The model learns to reconstruct masked values from context, forcing it to capture the statistical structure of methylation patterns. The loss decreases to 0.0013 over 250 epochs.

Table 5: BMFM-RNA pretraining results (MLM on β -values, 250 epochs).

Split	Loss	MSE	MAE	PCC
Train	0.00132	0.00132	0.0221	0.994
Validation	0.00147	0.00147	0.0234	0.994
Test	0.00133	0.00087	0.0195	0.997

The encoder achieves strong reconstruction performance: Test PCC (Pearson correlation) = 0.997 indicates the model accurately predicts masked β -values from surrounding context. The low test MAE (0.0195) on the $[0,1]$ scale demonstrates precise value reconstruction. Figure 1 shows the training dynamics. For detailed pretraining metrics, see: **BMFM-RNA Pretraining W&B**.

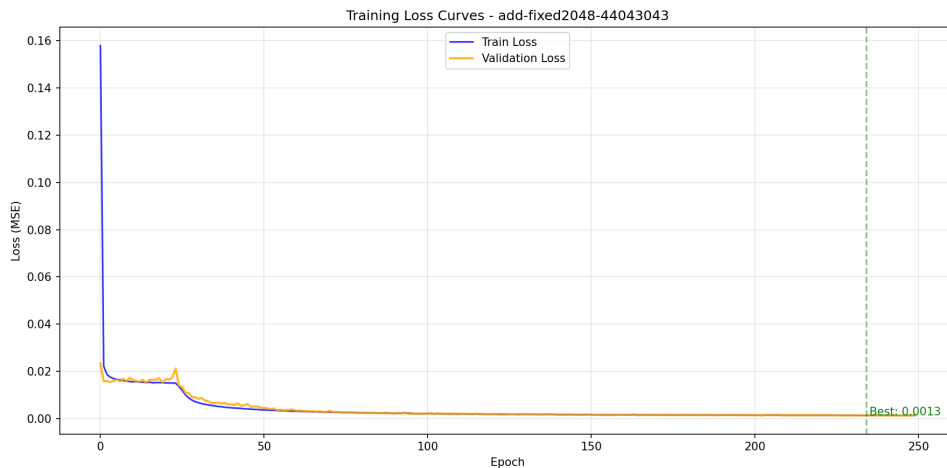


Figure 1: Pretraining loss curves for BMFM-RNA encoder (MLM objective). The model converges after approximately 150 epochs, achieving final test loss = 0.00133 and PCC = 0.997.

7.5 Fine-Tuning Pipeline

The pretrained encoder is fine-tuned for age regression:

- **Pooling:** Mean pooling over token outputs (skip CLS)
- **Age head:** MLP ($512 \rightarrow 256 \rightarrow 128 \rightarrow 1$) with LayerNorm, GELU, Dropout(0.2)
- **Freeze strategy:** Encoder frozen epochs 0–4, unfrozen epoch 5+ with $10\times$ lower LR
- **Loss:** MSE on z -score normalized ages

Table 6: Training hyperparameters (BMFM-RNA fine-tuning).

Hyperparameter	Value
Optimizer	AdamW
Learning rate	5×10^{-4}
Weight decay	0.01
LR schedule	Cosine decay with 200-step linear warmup
Batch size	32 (effective 64 with gradient accumulation $\times 2$)
Max epochs	300
Precision	16-bit mixed
Early stopping	Patience = 60 epochs on val/MAE
Checkpoint	Best validation MAE

7.6 Fine-Tuning Results

To assess generalization, we train on 5 different random seeds (40–44) and report mean $\pm$ std.

Table 7: BMFM-RNA fine-tuning results (5 seeds).

Metric	Mean $\pm$ Std
Validation MAE (years)	4.84 ± 0.16
Best Test MAE (years)	4.67
Test MedAE (years)	3.39
Test RMSE (years)	6.05
Test R^2	0.926 ± 0.005
Test PCC	0.965

Training run up to 250 epochs with early stopping. The model explains 92.6% of variance on the test set. Experiment tracking: **BMFM-RNA Fine-tuning W&B**.

8 Experimental Setup

We use the same 8K-CpG dataset as the MethylGPT baseline (11,453 samples, train/valid/test split). The mean prediction baseline ($\hat{y}_i = \bar{y}_{\text{train}}$) serves as a reference.

9 Final Comparison

Table 8: Comparison across all approaches on the test set.

Model	Test MAE	Test R^2
Mean prediction	22.82	0.00
MethylGPT baseline	4.95	0.911
BMFM-RNA (ours)	4.67	0.931

The BMFM-RNA model achieves 6% improvement in MAE (4.67 vs 4.95 years) and 2% improvement in R^2 (0.931 vs 0.911) over MethylGPT. Both models reduce error by approximately 78% compared to mean prediction (22.82 years).

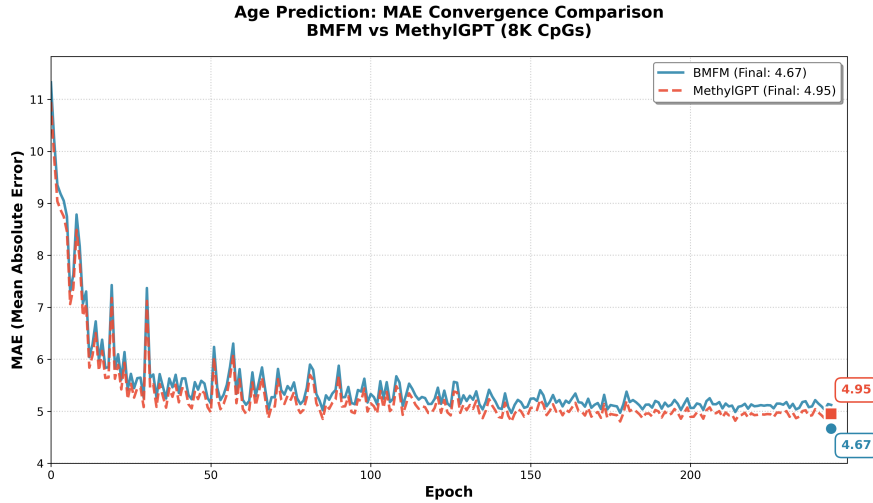


Figure 2: MAE comparison across models on the test set.

10 Discussion and Conclusions

- **Performance.** BMFM-RNA’s multi-field tokenization and larger capacity yield consistent improvements over MethylGPT (6% MAE reduction, 2% R^2 improvement). The explicit separation of site identity and methylation value appears beneficial for capturing age-related methylation patterns.
- **Key implementation details.** PyTorch Lightning’s `configure_optimizers()` is called once at initialization. If encoder parameters are frozen at init, they’re excluded from the optimizer. Fix: use `optimizer.add_param_group()` when unfreezing. Mean pooling over tokens works better than [CLS] for regression after MLM pretraining.
- **Overfitting.** Train MAE = 2.21 years vs test MAE = 4.67 years indicates moderate overfitting (train–test gap of ~ 2.5 years). Dropout(0.2) helped reduce this gap.
- **Capacity vs. gains.** Despite $10\times$ more parameters (23M vs 2M), improvement is 0.28 years, suggesting methylation age prediction may be approaching a performance ceiling.

- **Generalization.** Results across 5 random seeds show consistent performance (MAE: 4.84 ± 0.16 years), confirming model stability.

References

1. Ying, K., et al., “MethylGPT: a foundation model for the DNA methylome,” *bioRxiv*, 2024.
2. Dandala, B., et al., “BMFM-RNA: An Open Framework for Building Transcriptomic Foundation Models,” *arXiv*, 2025.
3. Horvath, S., “DNA methylation age of human tissues and cell types,” *Genome Biology*, 14(10):R115, 2013.